

**What makes an
asset the
dominant
intermediary?**



**Centre for Blockchain
Technologies**

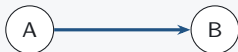
The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

How we reconstruct a route from one transaction

Direct swap

one transaction



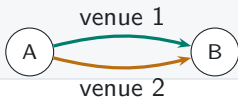
Sequential vehicle route

one transaction



Parallel split

one transaction



Round trip

one transaction



Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



Count-weighted dominance

Share of indirect routes that pass through k

Value-weighted dominance

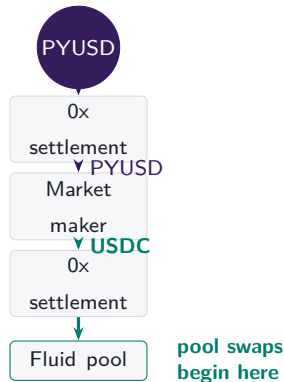
Share of comparable routed value that passes through k

The transaction starts with PYUSD

EXPLORER RECORD

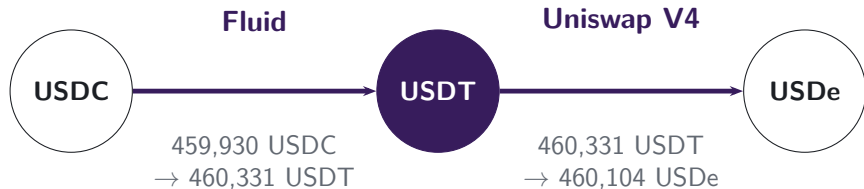
Transaction details 0x				
⚡ Swap 460.00K PYUSD for 460.10K USDE on 0x 0x 6 Explorers				
Details Token transfers User operations Internal txns Logs State Raw trace				
PayPal USD ERC-20 Transfer	0xf7...0c21 → Mainnetsettler	460,000 \$459,854.18		
PayPal USD ERC-20 Transfer	Mainnetsettler → Market Maker: 0x51c7.....	460,000 \$459,854.18		
USDC ERC-20 Transfer	Market Maker: 0x51c7..... → Mainnetsettler	459,929.80974 \$459,741.24		
USDC	Mainnetsettler → Fluid: Liquidity	459,929.80974		

SETTLEMENT HANDOFF



PYUSD reaches 0x settlement; the market maker supplies USDC for execution in the pools.

USDT is the intermediary in the connected pool route



vehicle

The connected component converts USDC into USDe across two pools.
USDT is the vehicle because it links the two legs.

0xbda1d07f...bf67ad9, 10 January 2026

Architecture changes routing opportunities

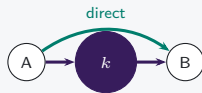
V1 mandatory hub



Only ETH-token pools: token-to-token execution must pass through ETH.

Hub use is imposed

V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

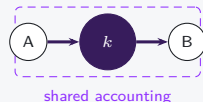
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



Two pools still execute; intermediate token movement may be netted.

Settlement and vehicle use can diverge

Architecture feasible rules

Exposure realised pool and route use

Calendar date timing only

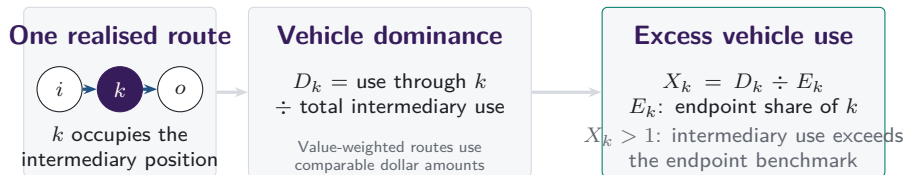
The sample spans Uniswap V1 through V4

DATA

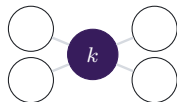


Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Vehicle dominance aggregates realised intermediary choices

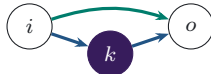


Separate characteristics of the asset and route set



Network position

position of k in the feasible graph

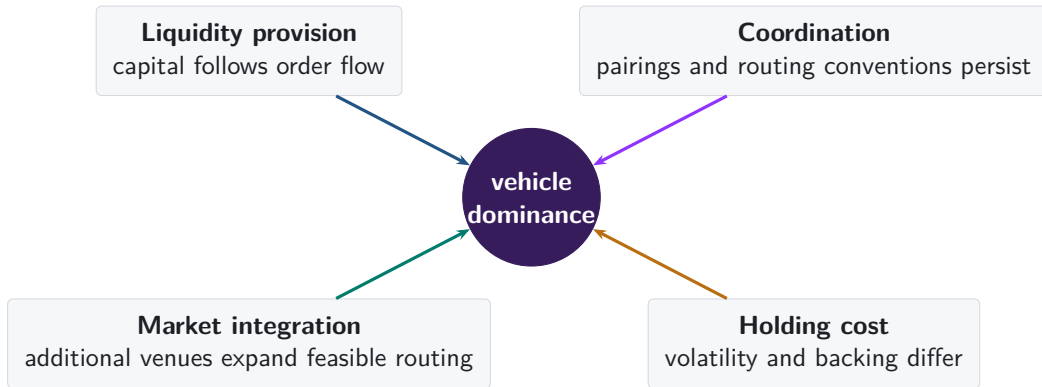


Execution cost

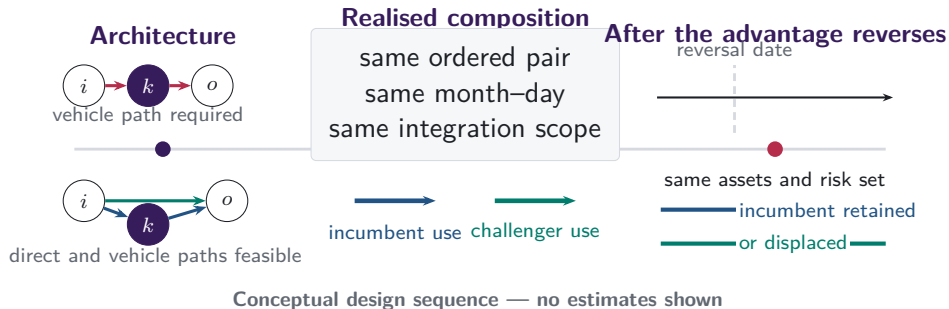
which route returns more at the same state

Endpoint demand supplies the benchmark; network position and same-state route costs remain separate characteristics.

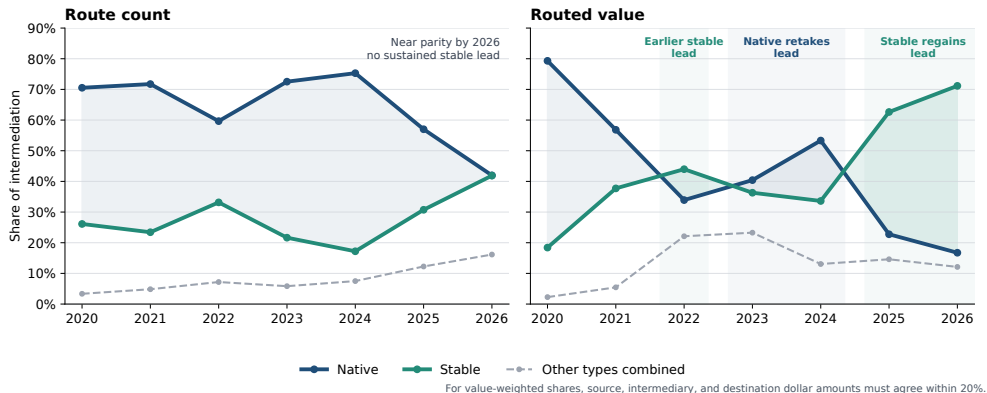
Why might one asset become the dominant vehicle?



Architecture opens routes; realised trades reveal which vehicle is used

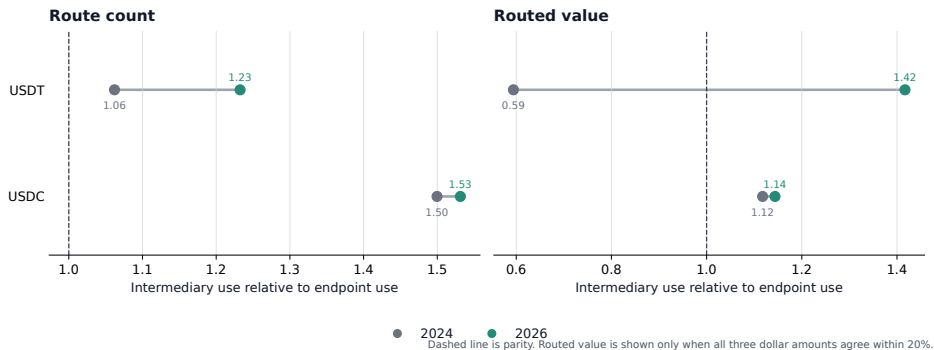


Stablecoins regain the lead in routed value



Annual shares of observed vehicle routes; the figure does not attribute the changes to protocol design.

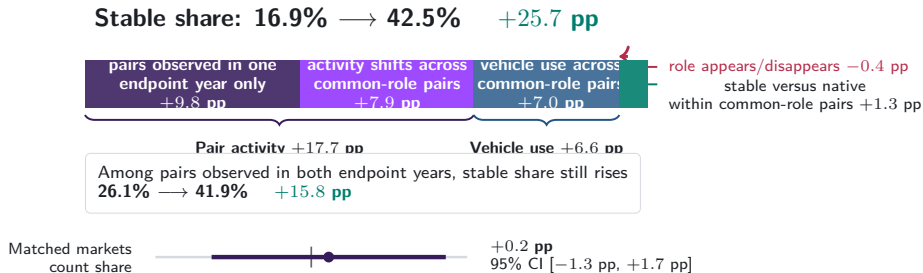
USDT drives the post-2024 rise in stablecoin intermediation



USDT crosses parity by routed value and extends its count advantage; USDC changes little. Routed value includes trades whose three dollar amounts agree within 20%.

93.7% of the count-share change and 85.6% of the routed-value change occur within the single- and cross-venue categories.

Pair activity and vehicle use: 94.9% of the gain

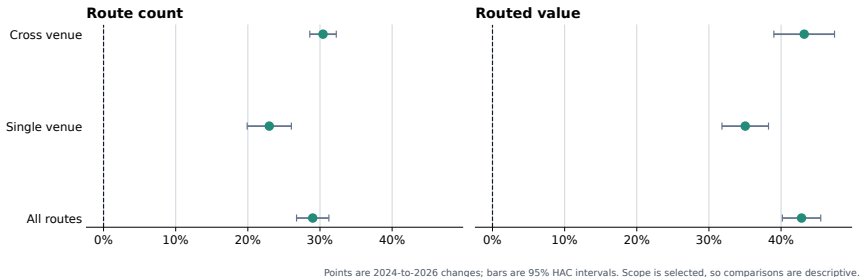


Dollar-weighted routes show the same broad allocation. Stable share rises +42.8 pp, with +26.2 pp from activity weights among continuing pairs and only 0.0 pp from stable-for-native substitution within them.

Route-count accounting for ordered endpoint pairs, pooling single- and cross-venue routes on days of the calendar year observed in both endpoint years through 30 June. Matched markets additionally hold the ordered endpoint pair and single- versus cross-venue route category fixed. Neither comparison holds venue sequence, feasible alternatives, trade size, liquidity, cost, or router state fixed.

Stablecoin use rises within each venue scope

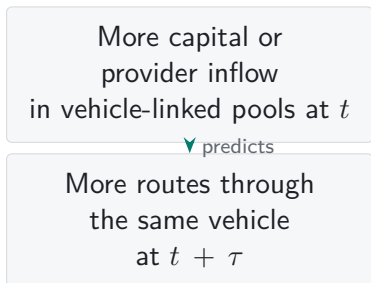
Stable share rises within realised route scopes



The increase appears within both realised route scopes, so a change in the single- versus cross-venue mix cannot account for the aggregate rotation. Cross-venue status remains selected; the comparison does not identify an effect of integration, routing software, or protocol architecture.

Does liquidity move before vehicle use or follow it?

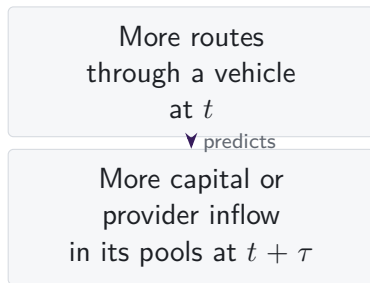
LIQUIDITY PRECEDES USE



Liquidity supply may help a vehicle attract later routes.

V2 reserve-based capital and V3 mint/burn flows are measured separately at exact calendar lags. Either direction may also reflect common demand, volatility, or routing shocks.

USE PRECEDES LIQUIDITY



Providers may instead follow routed demand.

V4 separates the economic route from physical settlement

ECONOMIC ROUTE



Pool calls route
through intermediary k .

V4 may net the token movements
used to settle those calls.

COMPARISON

- Compare routes in the same week with the same endpoints, intermediary, direction, and trade-size group.
- Record route use and physical token transfers separately.
- Compare feasible alternatives before drawing conclusions about vehicle choice.

The architecture can make a physical transfer unnecessary. The empirical test asks whether transfers fall among otherwise comparable routes.

Cheaper challengers gain share as incumbents lose it

30-day outcome	$\hat{\beta}$	SE	t
Challenger vehicle-share response	4.825	0.366	13.17
Incumbent vehicle-share response	-3.872	0.420	-9.22
Challenger overtakes incumbent	9.947	0.486	20.48

$N = 57,989$ endpoint-pair-intermediary observations; 2,213 date clusters. Exact alternating-projection absorption and CR1 rank correction.

The estimates are large in the small set of endpoint markets and intermediary assets we can follow. They remain descriptive because the panel does not compare each market against a fixed set of feasible routes.

Current architecture changes do not provide a clean before–after comparison

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	–	–	–
Immediate exit contrast	–	–	–

At every threshold, entries and exits coincide with changes in the comparison set or with the appearance and disappearance of vehicle roles. These windows therefore cannot separate an architecture effect from changes in which pairs and vehicles remain active.

A before–after estimate requires the same feasible alternatives before and after the architecture change.

Stablecoins regain the value lead as route counts approach parity

- Stablecoins lead routed value in 2022 and again in 2025–2026, but fall behind native assets in 2023–2024.
- USDT drives the post-2024 rise in excess stablecoin intermediation; USDC changes little.
- Pair activity and vehicle use account for almost all of the aggregate stablecoin gain. Their economic cause remains open.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

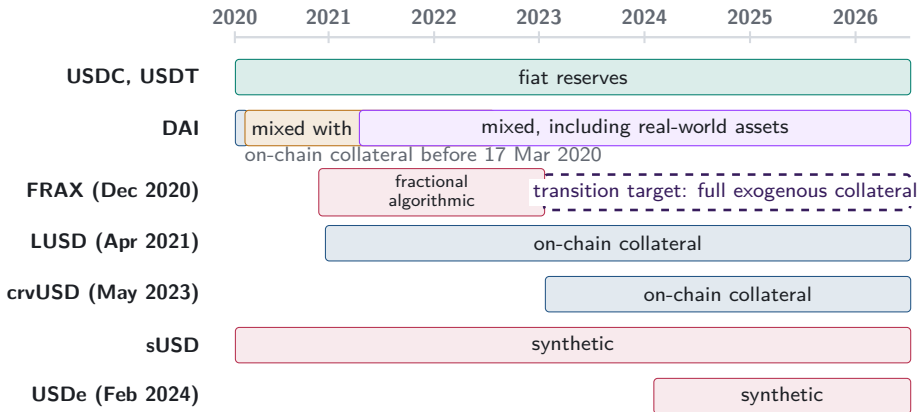
A1. One reconstructed route is one unit



one route unit
two swap legs

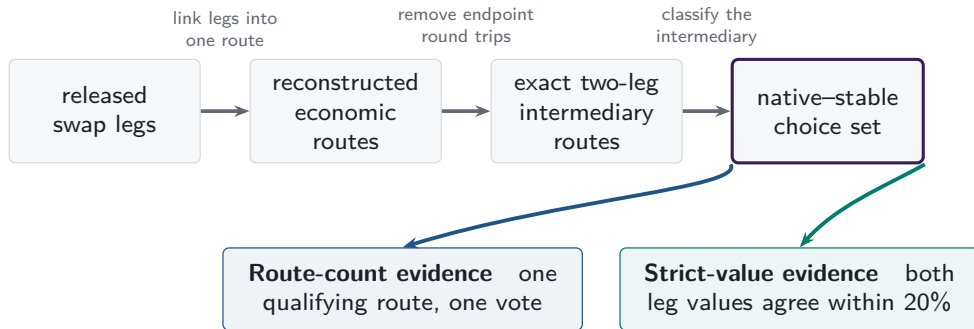
- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

A2. Stablecoin backing cannot be treated as fixed



Bars begin at public launch or the sample start for older tokens. DAI's segments mark the backing regimes admitted by the classification; the dashed FRAX arrow is a governance target, not an observed collateral share.

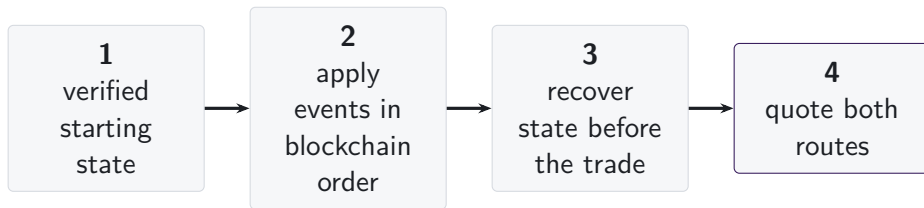
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

The common object is the state immediately before execution.
The fields required to recover that state vary with the pricing rule.

■ required state

□ hooks outside comparison

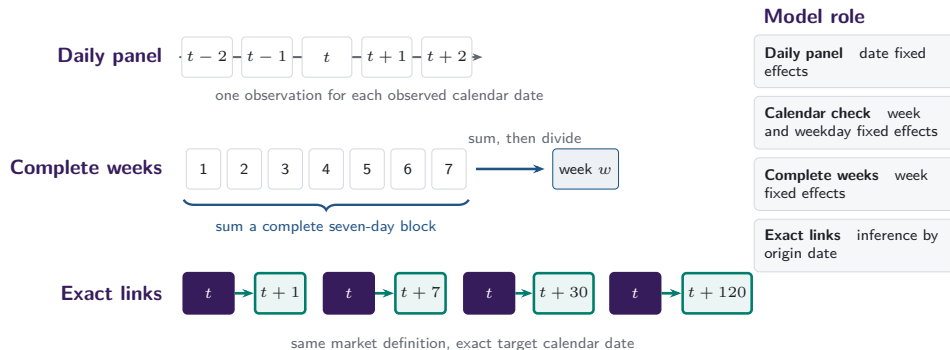
A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.



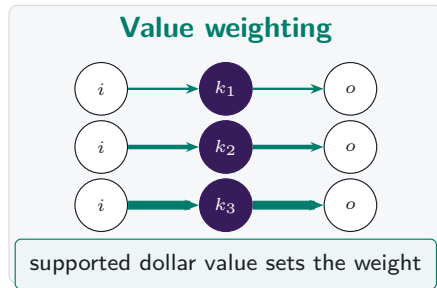
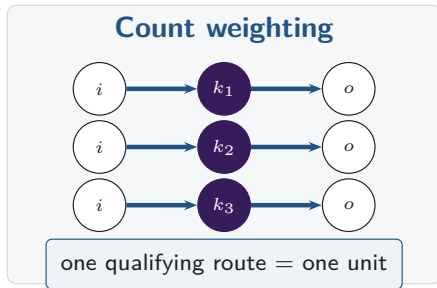
same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count and value define separate dominance margins



Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S_{k,t}^I = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

How often asset k sits inside a route.

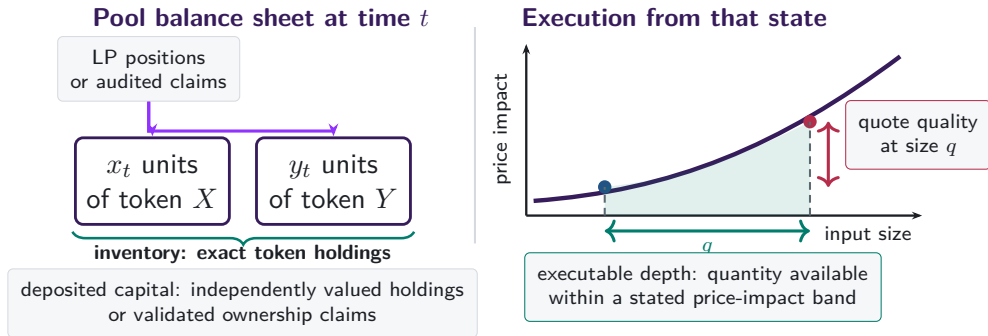
Endpoint share

$$S_{k,t}^E = \frac{N_{k,t}^{\text{in}} + N_{k,t}^{\text{out}}}{\sum_j N_{j,t}^{\text{in}} + N_{j,t}^{\text{out}}}$$

How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

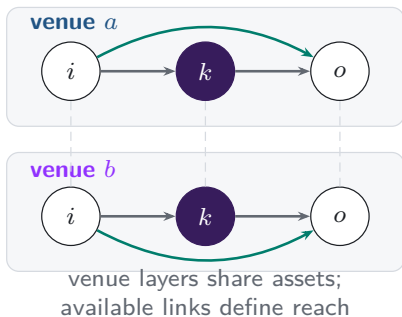
A10. Capital, inventory, and depth are different objects



Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen

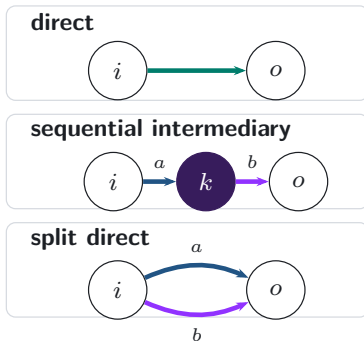
Feasible venue layers



realised
execution



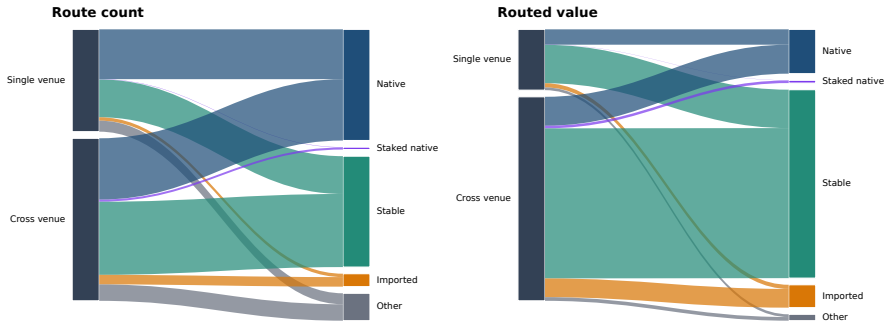
Realised route topology



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

The ribbons classify realised intermediation jointly by venue scope and intermediary type. They describe selected routes; they do not identify an effect of integration or the unchosen opportunity set.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
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- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
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